

a Coding example

A coding agent fixes a bug using execution feedback.

Task

Sum integers 1 through n.

$$S(n) = \sum_{i=1}^n i$$

Buggy program a^0

```
def sum_to(n):  
    return sum(range(1, n))
```



Coding agent

analyzes, repairs, revises

Corrected program a^1

```
def sum_to(n):  
    return sum(range(1, n + 1))
```

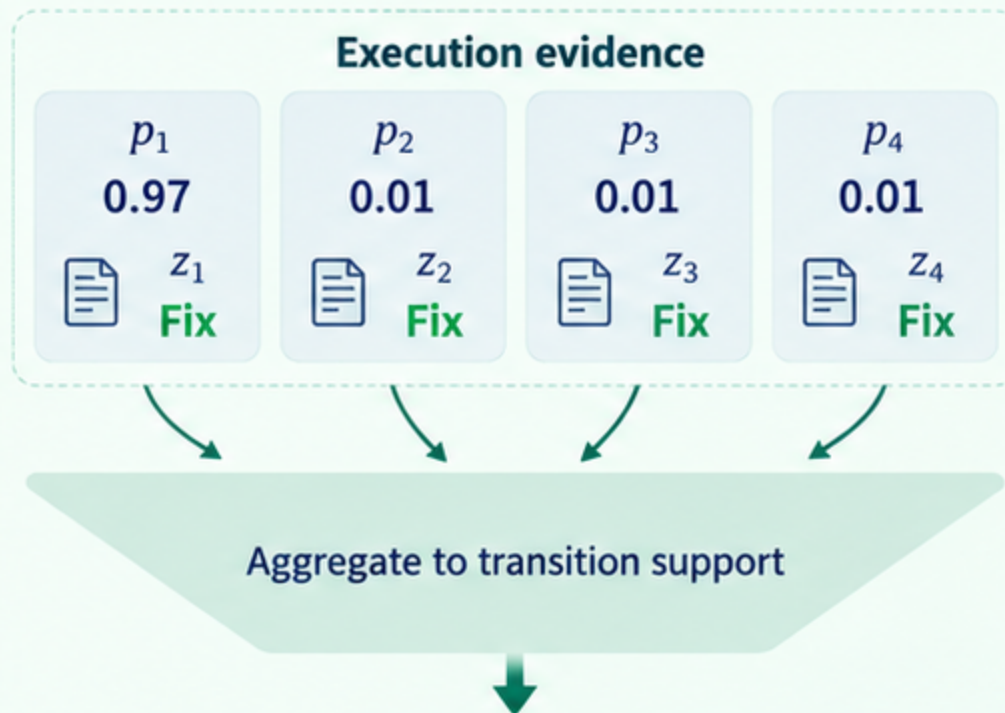
Public tests

n	Expected	Before (a^0)	After (a^1)
1	1	0 ✗	1 ✓
2	3	1 ✗	3 ✓
3	6	3 ✗	6 ✓
4	10	6 ✗	10 ✓

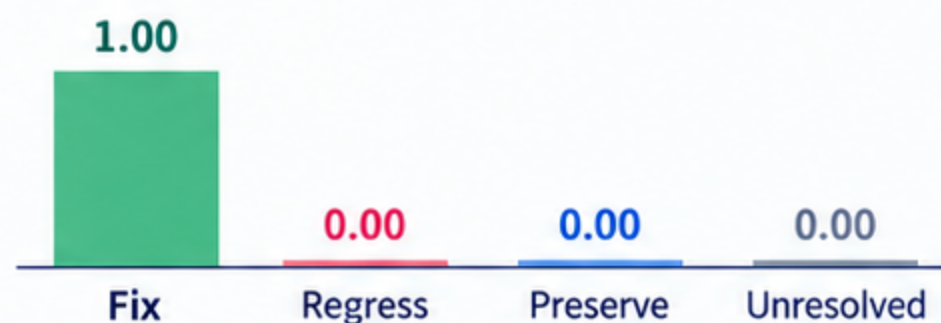
b Direction

direction

Relative support r from execution evidence.



Support over transitions r



Relative support r = which transition the evidence favors.

$$r = (1, 0, 0, 0)$$

c Strength

strength

The same direction r can come from different evidence concentration patterns.

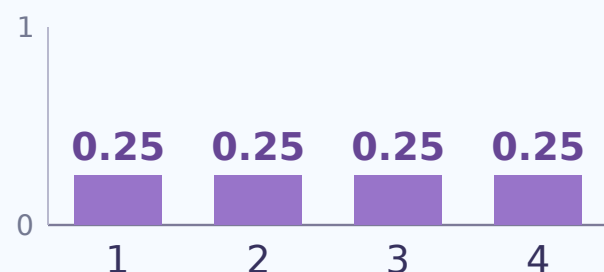
$$E(p) = \left(\sum_i p_i^2 \right)^{-1}$$

Effective evidence mass.

Broad relevance

$$p = (0.25, 0.25, 0.25, 0.25)$$

$$E(p) = 4.000$$



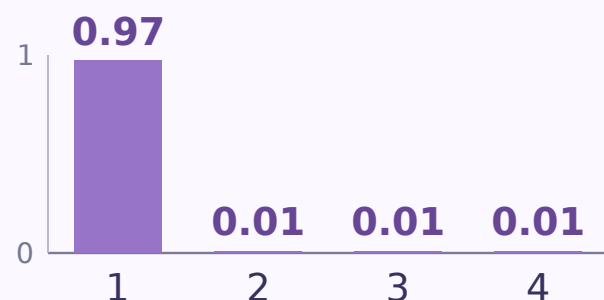
Supervised weight (same r)

Larger w

Concentrated relevance

$$p = (0.97, 0.01, 0.01, 0.01)$$

$$E(p) = 1.062$$



Supervised weight (same r)

Smaller w

Same support; different learning weight.

p : weights over executions 1-4
 r : support over four transition categories